

# Structural studies of phosphate glasses with high ionic conductivity

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## Abstract

Lithium phosphate glasses with different chemical composition have been studied using infrared absorption spectroscopy (FTIR) and photoelectron spectroscopy (XPS). FTIR and XPS spectra were analyzed to determine the main structural units present in the glass. Absorption bands were identified and assigned to appropriate bond vibrations. It has been shown that analyzed glasses are built up mainly of short chains and rings consisting of metaphosphate structural units –  $Q^2$ . Some pyro- and orthophosphate units  $Q^1$  and  $Q^0$ , respectively are also present. Photoelectron spectroscopy studies have revealed that titanium exists in two valence states: as  $Ti^{3+}$  and  $Ti^{4+}$ . Increasing  $TiO_2$  content results in shortening of P–O bonds but leads to elongation of P=O bonds in  $PO_4$  tetrahedra.

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## 1. Introduction

Phosphate glasses have been the object of recent interest because of their unique properties. They have a very high transmission in the ultraviolet region compared to silicate glasses. They are suitable materials for high power lasers because of low thermo-optical coefficient and large emission. More information can be found elsewhere [1]. Most biocompatible glasses are also based on phosphate glasses. Other applications are connected with a very low transition temperature of some alkali-alumino-phosphate glasses which indicates the possibility of connection with organic polymers [2]. Phosphate glasses bearing lithium have relatively high ionic conductivity, thus they can be applied in energy storage devices and solid state batteries [3–5].

The structure of phosphate glasses is formed by  $PO_4$  tetrahedra connected by corners similar to the silica structure. Vitreous  $P_2O_5$  has a three-dimensional network build up with  $PO_4$  tetrahedra connected by three of their four corners. The fourth one is occupied by terminal double-bonded oxygen. Introducing of glass network modifiers, RO or  $R_2O$  results in breaking P–O–P bonds, and terminal oxygens are formed. Depending

on the number of the bridging oxygen the phosphate tetrahedra can be described as  $Q^i$ , where ‘i’ represents the number of the bridging oxygens and can have a value of 0, 1, 2, 3. Such notation was first used by Lippma [6], and is widely accepted in the literature for describing structures of phosphates and silicates. A framework built up with  $Q^3$  denotes a fully polymerized structure, while  $Q^2$  unit gives only a two-dimensional structure based on chains or rings.  $Q^1$  means two tetrahedra connected by a corner and  $Q^0$  means isolated tetrahedra.  $Q^1$  and  $Q^0$  units are characteristic for orthophosphate structures [7].

The main goal of this work was to identify the main structural units of the studied glasses, and to explain the role of  $SiO_4$  tetrahedra and  $TiO_4$  polyhedra in structure formation.

## 2. Experimental procedure

Glasses were prepared by the conventional melting method. Baths were prepared with chemical pure reagents:  $Li_2CO_3$ ,  $NH_4H_2PO_4$ ,  $SiO_2$ ,  $Al_2O_3$ , and  $TiO_2$ . The chemical compositions of the studied glasses are shown in Table 1. Compositions of these glasses correspond to glasses having high ionic conductivity [8,9]. Synthesis was carried out in an alumina crucible in air. The crucible was covered to avoid volatilization losses.

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Table 1  
Chemical compositions of studied glasses

| Symbol | Chemical composition (mol%) |                               |                   |                                |                  |
|--------|-----------------------------|-------------------------------|-------------------|--------------------------------|------------------|
|        | SiO <sub>2</sub>            | P <sub>2</sub> O <sub>5</sub> | Li <sub>2</sub> O | Al <sub>2</sub> O <sub>3</sub> | TiO <sub>2</sub> |
| T1     | 21.5                        | 48                            | 26                | 4.5                            | 0                |
| T2     | 13.5                        | 46                            | 26                | 4.5                            | 10               |
| T3     | 11.5                        | 41                            | 26                | 4.5                            | 17               |
| T4     | 10.5                        | 40                            | 26                | 4.5                            | 19               |
| S1     | 12.5                        | 42                            | 32.5              | 4.5                            | 8.5              |
| S2     | 14.5                        | 42                            | 30.5              | 4.5                            | 8.5              |
| S3     | 17.5                        | 42                            | 27.5              | 4.5                            | 8.5              |

Error due to volatilization and crucible corrosion was corrected and was lower than 0.5 mol%.

Glasses were melted at 1473 K during 30 min. The melting conditions were kept constant for all glasses because these glasses are very sensitive to preparation procedure. The melt was next poured out on stainless-steel plate and 3 mm thick samples were formed. Glasses with higher amount of TiO<sub>2</sub>-exceeding 15 mol% were prepared by pouring the melts into the container filled with liquid nitrogen. This method allowed obtaining glassy samples even in case of glasses with high tendency to crystallization. Glass plates were next annealed during 2 h at temperature 10 °C below  $T_g$ . The annealing temperature was established on DTA measurements. Cold samples were kept in an exsiccator to avoid water vapour attack. Chemical compositions of the samples were examined by chemical analyzes and, if necessary, composition of the bath was corrected to obtain glass with the designed composition. X-ray diffraction method was used to confirm absence of crystallization. IR spectroscopic analyzes were carried out with a fast

Fourier transform IR spectrometer. The absorption spectra were measured using standard KBr pellet technique from 400 to 4000 cm<sup>-1</sup> with a resolution of 4 cm<sup>-1</sup>. X-ray photoelectron spectroscopy analyzes were performed on ‘as made’ samples using a spectrometer with hemispherical analyzer. Spectra were obtained using Al K $\alpha$  radiation source operated at 200 W and 10 keV ( $h\nu$  1486.6 eV). Electron energy analyzer was set to FAT mode with pass energy 22 eV. The shift of the binding energy due to surface charging effect was calibrated by assuming binding energy of C 1s to be always 284.6 eV. Spectra were fitted using a computer software. The binding energies were accurate to  $\pm 0.2$  eV. The sample was not grounded and C 1s spectra were recorded twice, at the beginning and at the end of each series of measurement. Two series of glasses with different amount of titanium, and silica were selected to establish the influence of these oxides on the structure of phosphate glasses.

### 3. Results

MIR spectra of phosphate glasses generally consist of four groups of bands: 1300–1200, 1100–900, 800–700 and 600–400 cm<sup>-1</sup> (Figs. 1 and 2). There is an agreement in literature that the first group of bands originate from P=O double bonds vibrations. Intensity of bands at  $\sim 1300$  cm<sup>-1</sup> strongly depends on phosphate content (Figs. 1 and 2). There are at least two bands in this region which is clearly seen in Fig. 2(a), (b). These two bands are connected with P=O vibrations affected by the presence of Ti atom located near this bond and

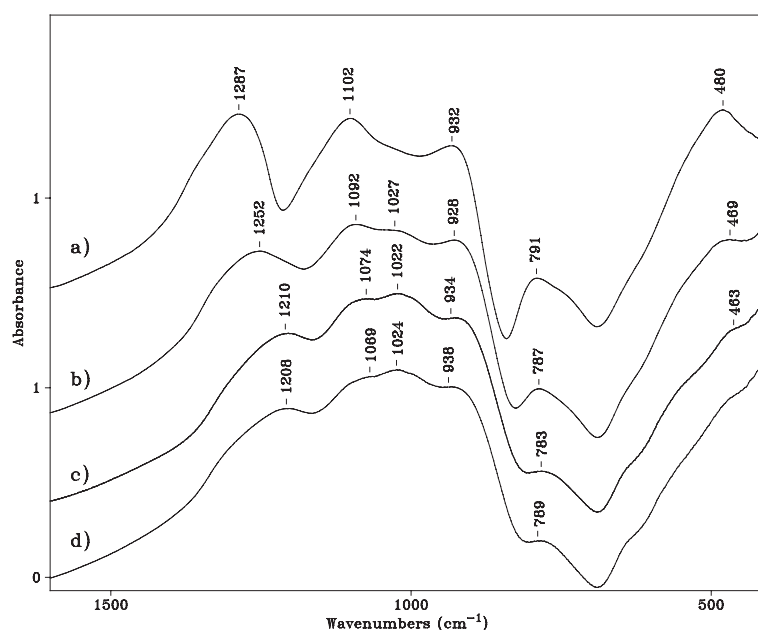


Fig. 1. MIR spectra of ‘T’ series glasses: (a) T1, (b) T2, (c) T3 and (d) T4.

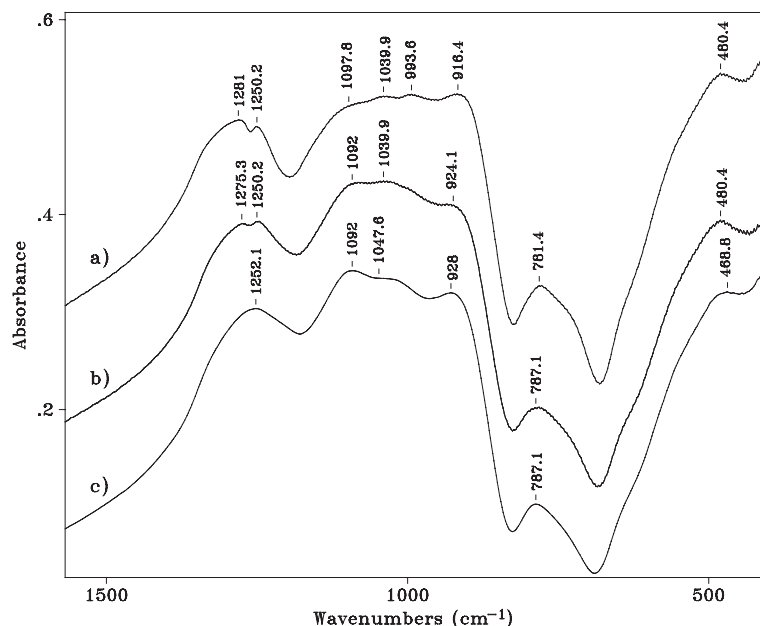


Fig. 2. IR spectra of 'S' series glasses: (a) S1, (b) S2 and (c) S3.

asymmetric stretching vibrations of O–P–O bonds [10]. Position of these bands strongly depends on the chemical composition of the glass. Titanium introduced into the glass with composition T1 leads to shifting of the bands at  $\sim 1290\text{ cm}^{-1}$  to lower frequencies  $\sim 1200\text{ cm}^{-1}$  (Fig. 1(a)–(d)). It means elongation of P=O and O–P–O bonds. It is reasonable to assume that titanium is located near P=O bond and some phosphate tetrahedra are connected through Si atoms forming P–O–Si bridges [11]. Lowering  $\text{SiO}_2$  content in favor of titanium leads to lowering of the number of P–O–Si bridges and  $Q^1$  type structural elements are formed. Band at  $\sim 1100\text{ cm}^{-1}$  results from the superimposition of the bands originating from asymmetric stretching Si–O bonds vibrations and symmetric P–O bonds vibrations in phosphate tetrahedra. Clear correlation between  $\text{SiO}_2$  content and this band is seen in Figs. 1 and 2. Absorption band at  $\sim 1020\text{ cm}^{-1}$  is connected with stretching vibrations of terminal oxygen P–O $^-$ , whereas the band at  $\sim 920\text{ cm}^{-1}$  with asymmetric vibrations of P–O–P bridges.

Bands from the range  $850\text{--}600\text{ cm}^{-1}$  are connected with symmetric stretching vibrations of P–O–P and Si–O–P bridges [11]. The origin of the bands at lower wavenumbers, below  $600\text{ cm}^{-1}$ , is not quite clear. It is usually assumed that in this range bands originate from vibrations of large groups of atoms, like 3,4-member rings or short chains. Superposing of bands due to vibrations of structural forms, mentioned above, as well as the bending vibrations of P–O–P and P–O–Si bridges give continuous spectrum usually observed in this region. Tetrahedrally co-ordinated lithium ion usually gives absorption band at the edge of MIR range  $500\text{--}400\text{ cm}^{-1}$  and in the far infrared region [12,13].

In the studied glasses titanium is present in the form of  $\text{Ti}^{4+}$  and  $\text{Ti}^{3+}$ . The amount of  $\text{Ti}^{3+}$  established on the basis of photoelectron spectroscopy is about 30% and it strongly depends on the melting conditions. More oxidizing conditions lead to higher  $\text{Ti}^{4+}$  content. The region of titanium for a sample with the composition T2 is shown in Fig. 3. Titanium with oxidation number 3+

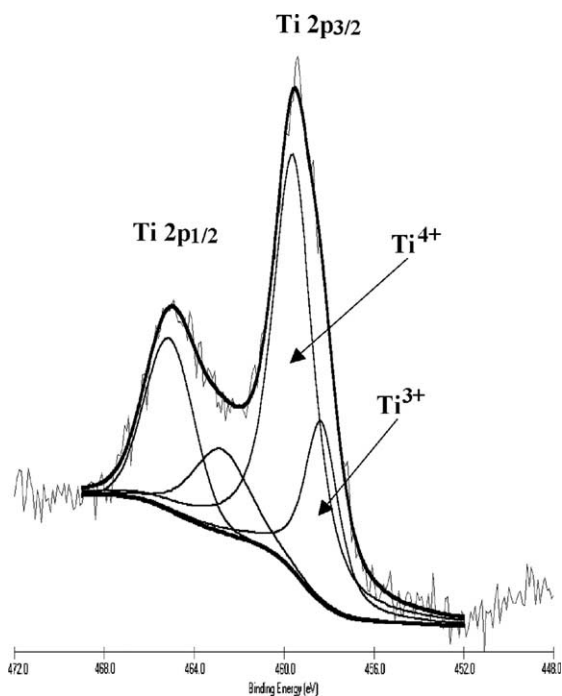


Fig. 3. XPS spectra of Ti 2p region. Fitting procedure reveals two peaks assigned to  $\text{Ti}^{3+}$  and  $\text{Ti}^{4+}$ .

has lower binding energy, amounting to 458.3 eV for  $2p_{3/2}$  peak. Binding energy of  $Ti^{3+}$  observed in this study is about 1 eV higher than that reported for  $Ti_2O_3$  – 457.2 eV [14]. Higher binding energy is due to high ionic character of Ti bonds in phosphate glasses and confirms the localization of Ti atoms near the  $P=O$  bond. Coordination number of titanium  $Ti^{3+}$  is 6 and 5 or 4 + 1 [15,16]. Notation 4 + 1 is used to emphasize the deformation of the coordination polyhedron. The second peak with the position 459.6 eV ( $2p_{2/3}$ ) is connected with  $Ti^{4+}$  (Fig. 3). Similar binding energy for titanium  $Ti^{4+}$  was measured in case of aluminophosphate glasses bearing lithium and titanium [14]. The binding energy of  $Ti^{4+}$  is higher than for example in  $TiO_2$  [17] and it reflects a more ionic character of Ti–O bonds in phosphate glasses than in  $TiO_2$  oxide. Titanium  $Ti^{4+}$  coordinated by four oxygen acts as glass-former oxide, while  $Ti^{3+}$  is a modifier of the glass structure.

The region of oxygen contains two peaks originating from oxygen having different chemical environment (Fig. 4). The peak with lower binding energy – 530.6 eV comes from terminal oxygen bonded with phosphate and lithium or titanium  $Ti^{3+}$ . The bridging oxygen from P–O–P and P–O–Si bridges has higher binding energy – 532.2 eV. Most of oxygen, about 60%, is present in the form of non-bridging oxygen (Fig. 4). This is in agreement with the chemical character of the studied glasses – ‘invert glasses’ with  $O/P \approx 3.5$ . The skeleton of such glasses is build up of short chains (2,3 tetrahedrons) or rings connected with each other through the modifier

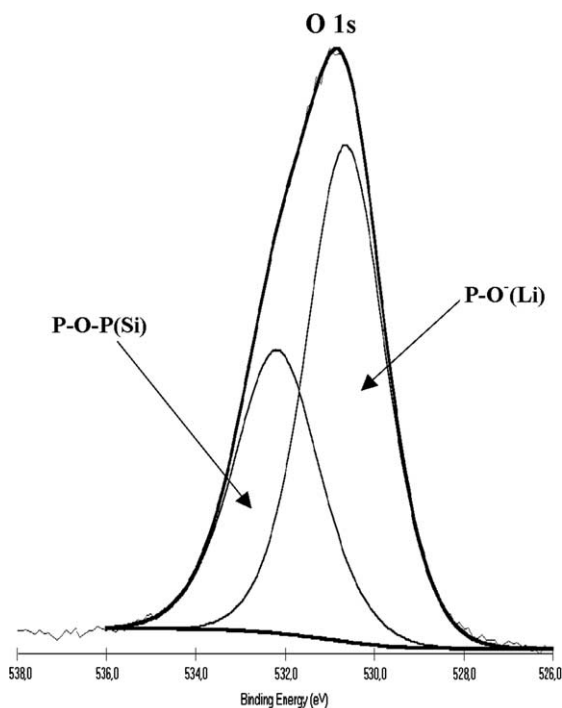


Fig. 4. XPS of O 1s region. Non-bridging oxygen peak is much higher than the peak connected with the bridging oxygen.

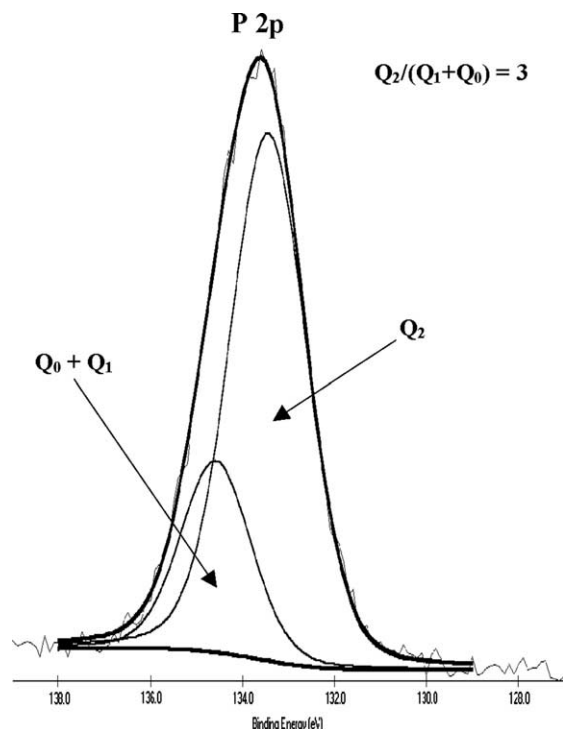


Fig. 5. XPS results, P 2p region. Two different phosphate sites can be distinguished after fitting procedure.

cations –  $Ti^{3+}$ ,  $Al^{3+}$ . Very similar results have been found by NMR spectroscopy in case of sodium phosphate glasses [18]. Some structural elements, characteristic for pyro- and orthophosphates –  $Q^1$  and  $Q^0$ , leading to the observed high number of terminal oxygen, could also be expected. However, because an XPS is a very surface sensitive technique (only few monolayers can be analyzed), the number of the bridging oxygen far below the surface can be expected to be higher. On the basis of the analysis of P 2p region (Fig. 5), it is possible to distinguish phosphate coming from  $Q^2$  groups and from pyro- and orthophosphate groups  $Q^1$  and  $Q^0$ . Phosphate from metaphosphate groups –  $Q^2$  has lower binding energy – 133.4 eV whereas, that from pyro- and orthophosphate groups about 1 eV higher – 134.5 eV [19]. The amount of  $Q^0$  units is governed, according to van Wazer's theory [20], by the equilibrium:  $Q^0 \rightleftharpoons 2Q^1 - Q^2$ . The binding energy of lithium Li 1s is 55.0 eV and such a value is observed in case of other lithium glasses [16], but high value of FWHM (full width at half maximum) parameter indicates differentiation of lithium sites in the glass structure.

#### 4. Discussion

Studied glasses belong to the class of ‘invert’ glasses so three dimension structure is build up with a help of modifier cations (titanium and alumina in this case).

XPS analyzes confirmed that the titanium is present in the glass structure in the form of  $Ti^{4+}$  and  $Ti^{3+}$ . The amount of  $Ti^{4+}$  strongly depends on the melting conditions (oxidizing or reducing) and on the overall amount of titanium in the glass. Some amount of titanium is always present in the form of  $Ti^{3+}$  even if the glasses were melted in ambient or oxidizing conditions [16,21]. In our case, amount of  $Ti^{3+}$  determined from the XPS measurement is in the range of 20–40% depends on a total amount of titanium in the glass composition.  $Ti^{4+}$  ions act as a glass forming while  $Ti^{3+}$  ions are glass modifier.  $Ti^{3+}$  ion is located near  $P=O$  double bonds and is coordinated by four oxygen. This position is usually occupied by glass modifier ion [11,23].  $Ti^{4+}$  ions connect  $PO_4$  tetrahedra through non-bridging oxygen. Such location of  $Ti^{4+}$  is supported by XPS (O 1s region) results. Increase of total amount of titanium in the glass decrease number of non-bridging oxygen. Phosphate network is build up with  $Q^2$  and  $Q^1$  groups.  $Q^2$  units form short chains (up to several  $PO_4$  tetrahedra) terminated by  $Q^1$  unit via modifying cations. Similar structure was proposed for binary ultraphosphate glasses [11,22–24]. Some  $Q^0$  units are also expected especially near surface region.  $Q^2$  to  $Q^1 + Q^0$  ratio, estimated from XPS is equal 3 to 1 what is consistent with literature [25–27]. Silica and alumina are incorporated into the phosphate network and form three-dimensional skeleton by connecting  $Q^2$  chains and/or rings.

## 5. Conclusions

- The studied glasses belong to pyrophosphate glasses with  $O/P \approx 3.5$ .
- The structure is build up mainly with short chains or rings based on  $Q^2$  and  $Q^1$  units. Silica is incorporated into the phosphate structure through Si–O–P and Si–O–Al(P) bridges.
- The structure is not continuous in the sense of Zachariasen theory so these glasses belong to the group of ‘invert glasses’, where the structure is build up with smaller fragments of glass-forming ions linked through typical modifiers  $TiO_2$ ,  $Al_2O_3$ . Electric charge is balanced by lithium cations.  $Q^0$  units are also present.
- Titanium is present in the form of  $Ti^{4+}$  and  $Ti^{3+}$  but the amount of  $Ti^{3+}$  depends on the overall amount of titanium in the glass. The amount of  $Ti^{3+}$  in glasses containing more than 10 mol% is about 30%.
- The number of non-bridging oxygen considerably exceeds the number of the bridging oxygen what confirms that glass network is non-continuous.

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